MIL-S-50419A(MI)
12 July 1979
SUPERSEDING
MIL-S-50419(MU)
15 June 1970

MILITARY SPECIFICATION

SPACER, FOR 2.75 INCH ROCKET, MARK 40 and MARK 4

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawing for a type of metal spacer designed as spacer, for rocket motor Mark 40 and Mark 4.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition and Special Weapons, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-ESD (R&D), Redstone Arsenal, AL 35809, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1340

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DRAWINGS

ARMY

9243469 Spacer

* (Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

* 3.1 First article inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract. (See 6.4)

3.2 Material. Materials and parts shall be in accordance with the specified drawings and specifications.

* 3.3 Design and construction. The design and construction of the spacer shall be in accordance with Drawing 9243469 and with requirements specified herein.

3.4 Workmanship. All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, metal shavings, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing bare material. All required markings shall be neat and sharply defined. Known item defects shall not be submitted with the lot.

4. QUALITY ASSURANCE PROVISIONS

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* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of specification MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

* 4.2 First article inspection.

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* 4.2.1 First article sample. The contractor shall submit a First Article Sample to a Government approved facility as specified in the contract for evaluation in accordance with the provisions of 4.2.2. The First Article sample shall consist of thirty assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials (see 4.1.1.c). A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 of this specification and any or all requirements of the drawings.

4.2.3 Rejection. If any assembly or component fails to comply with any of the requirements and inspections, the First Article Sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and/or submit a new First Article Sample or portion thereof. Until a First Article Sample is accepted the contractor is, in no way authorized by the Government to initiate regular production unless directed by the Contracting Officer.

4.3 Inspection Provisions.

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4.3.1 Lot formation. The term "lot" as used in this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification or drawing not affecting safety, performance, interchangability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homegeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Inspection lots for the inspection of the item being procured and deliverable under the contract shall comply with MIL-STD-105 and shall be numbered with a unique lot number which shall be in accordance with MIL-STD-1168. Unless otherwise approved by the Contracting Officer, the inspection lot size of the item being procured and deliverable under the contract shall not exceed 40,000 units nor be less than 10,000 units. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor and the Government representative.

4.3.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.3.

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4.3.2.1 Spacer (see dwg. 9243469).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Outside diameter, max	Gage	01001
102.	Inside diameter	Gage	01002
103.	Outside diameter, including protrusion, max	Gage	01003
104.	Radius of protrusion	Gage	01004
105.	Thickness of material	Gage	01005
106.	Flatness	Gage	01006
107.	Parallelism	Gage	01007
Minor:	AQL 1.5 percent		
201.	Width, max	Gage	01008
202.	Protective finish missing or improper	Visual	01009
203.	Evidence of poor workmanship (see 3.4)	Visual	01010

* 4.3.3 Inspection equipment. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

* 4.3.3.1 Government rights to documentation. Inspection equipment drawings provided and revised in accordance with the requirements of this specification may be used by DoD activities for design, procurement, manufacture, testing, evaulation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

* 5. PACKAGING

5.1 Packing, packaging and marking. The metal spacers shall be packed and packaged into a suitable commercially available fiberboard container. The container shall be of a type which will assure arrival of its contents at the destination in an undamaged condition. Marking shall be in accordance with MIL-STD-129 and shall include the manufacturer's name and contract number.

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6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

* 6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. All submitted designs shall contain a reference to the applicable 5 digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. These code numbers are used to correlate the characteristics with the inspections listed in this specification. They shall also be cited as referenced on drawings of equipment designs submitted by the contractor (see 6.3.1).

* 6.3 Contractor design. Special calibration and test equipment necessary to accomplish the requirements as stated herein may be designed and manufactured by the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibrating instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted Contractor designs shall conform to DOD-D-1000, LEVEL 2.

* 6.3.1 Submission of contractor design. All submitted designs shall contain a reference to the applicable paragraphs of this specification, and test equipment shall be approved by the Procuring Activity prior to submission of first article sample. Partial submission of design is permissible. However, the completion date for design review will be based on the date of the final submission. Design review at the Procuring Activity will be accomplished approximately 30 days after receipt.

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* 6.4 Submission of first article samples. Instruction as to the location for the evaluation of the first article sample shall be obtained from the Contracting Officer. All inquiries should be forwarded to the address specified in the contract.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the past previous issue.

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Preparing Activity:
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Project No. 1340-A460

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USA Msl Command
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-50419A(MI) SPACER, FOR 2.75 INCH ROCKET, MARK 40 and MARK 4

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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